

ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Experiment 14: House Price Prediction using Linear Regression

Aim

To implement a House Price Prediction model using an appropriate machine learning algorithm and predict house prices based on different features.

Procedure

1. Import the required libraries.
2. Create a house price dataset.
3. Separate input features and target price.
4. Split the dataset into training and testing data.
5. Train a Linear Regression model.
6. Predict house prices using the test data.
7. Evaluate the model using MSE and R^2 score.
8. Predict the price of a new house.

Algorithm

1. Start.
2. Load the house price dataset.
3. Select features such as area, bedrooms, bathrooms and age.
4. Split the data into training and testing sets.
5. Train the Linear Regression model.
6. Predict house prices for the test data.
7. Calculate MSE and R^2 score.
8. Predict the price of a new house.
9. Display the results.
10. Stop.

Dataset 1 — Basic House Features

Area_sqft	Bedrooms	Bathrooms	Age_Years	House_Price
800	2	1	15	1800000
1000	2	2	12	2300000
1200	3	2	10	2800000
1500	3	2	7	3500000
1800	4	3	5	4300000

Dataset 2 — Location and House Features

Area_sqft	Bedrooms	Distance_City_km	Parking_Spaces	House_Price
750	2	12	1	1700000
950	2	10	1	2200000
1200	3	8	1	2900000
1450	3	6	2	3600000
1750	4	4	2	4500000

Dataset 3 — Modern House Features

Area_sqft	Bedrooms	Bathrooms	Floors	House_Price
900	2	1	1	2000000
1100	2	2	1	2500000
1300	3	2	1	3100000
1600	3	2	2	3800000
1900	4	3	2	4700000

Dataset 4 — Property Condition

Area_sqft	Bedrooms	Age_Years	Condition_Score	House_Price
850	2	20	5	1600000
1050	2	15	6	2200000
1250	3	12	7	2900000
1500	3	8	8	3700000
1850	4	5	9	4700000

Dataset 5 — Luxury House Features

Area_sqft	Bedrooms	Bathrooms	Garage_Capacity	House_Price
1500	3	2	1	4500000
1800	3	2	2	5500000
2200	4	3	2	6800000
2600	4	3	3	8200000
3000	5	4	3	9800000

Code

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X = np.array([
    [1200, 2, 1, 10],
    [1500, 3, 2, 5],
    [1800, 3, 2, 3],
    [1000, 2, 1, 15],
    [2200, 4, 3, 2],
    [1300, 2, 2, 8],
    [2000, 4, 2, 5],
    [1600, 3, 2, 7],
    [2500, 4, 3, 1],
    [1100, 2, 1, 12],
    [1900, 3, 2, 4],
    [2300, 4, 3, 3]
])

y = np.array([
    4500000, 6000000, 7500000, 3500000,
    9500000, 5000000, 8000000, 6500000,
    11000000, 4000000, 7800000, 10000000
])

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Prices:")
print(y_test)

print("Predicted Prices:")
print(y_pred)

print("Mean Squared Error:", mean_squared_error(y_test, y_pred))

print("R2 Score:", r2_score(y_test, y_pred))

new_house = [[1700, 3, 2, 5]]

prediction = model.predict(new_house)
```

Output

```
*** Actual Prices:
[7800000 4000000 4500000]
Predicted Prices:
[7931963.1028292  3907230.85494955 4401751.99653315]
Mean Squared Error: 11891014988.974337
R2 Score: 0.9958162965245986
Predicted House Price: 6932334.550857427
```

Result

The House Price Prediction model was successfully implemented using Linear Regression. The model predicts house prices based on area, number of bedrooms, bathrooms, and house age, and its performance was evaluated using MSE and R^2 score.